Algebra

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Equalities

Mastery of Elementary Algebra (a.k.a. "College Algebra") is a prerequisite for calculus, which is in turn a prerequisite for most statistics and data science courses. Nevertheless, each year, some students are still uncomfortable with algebraic manipulations of mathematical formulas. Therefore, I include this section as a quick reference.

Definition 0.1 (Transitive relation)**.** Let $\sim$ be a relation on a set S. The relation $\sim$ is **transitive** if, for all $a, b, c \in S$, $a \sim b$ and $b \sim c$ together imply $a \sim c$.

Example 0.1 (A transitive relation and one that is not)**.**

- $\leq$ is transitive: for example, $1 \leq 2$ and $2 \leq 5$, and indeed $1 \leq 5$.
- $\neq$ is not transitive: $1 \neq 2$ and $2 \neq 1$, but $1 \neq 1$ is false.

Theorem 0.1 (Equalities are transitive)**.** *If $a = b$ and $b = c$, then $a = c$. That is, $=$ is a transitive relation.*

Theorem 0.2 (Substituting equivalent expressions)**.** *If $a = b$, then for any function $f(x)$, $f(a) = f(b)$*

Definition 0.2 (Identity)**.** An **identity** is an equation between two expressions that holds for every value of its variables in a stated set, such as the real numbers.

Example 0.2 (An identity, and an equation that is not one)**.**

- $2(x + 1) = 2x + 2$ is an identity: it holds for every real x. For example, at $x = 3$, both sides equal 8.

- $x + 1 = 3$ is not an identity: it holds at $x = 2$, but at $x = 0$ the left side is 1 and the right side is 3.

Inequalities

Definition 0.3 (Strict and non-strict inequalities)**.** For real numbers a and b, $a < b$ ("a is less than b") and $a > b$ ("a is greater than b") are **strict inequalities**. The **non-strict inequalities** also allow equality:

- $a \leq b$ means $a < b$ or $a = b$;
- $a \geq b$ means $a > b$ or $a = b$.

Example 0.3 (Strict and non-strict inequalities)**.**

- $3 \leq 3$ is true, because $3 = 3$, but $3 < 3$ is false.
- $2 < 3$ and $2 \leq 3$ are both true.
- $4 \geq 5$ is false, because $4 > 5$ and $4 = 5$ are both false.

Theorem 0.3 (Adding to both sides of an inequality)**.** *If $a < b$, then $a + c < b + c$*

Theorem 0.4 (Negating both sides of an inequality)**.** *If $a < b$, then: $-a > -b$*

Theorem 0.5 (Multiplying both sides of an inequality by a positive number)**.** *If $a < b$ and $c > 0$, then $ca < cb$.*

Theorem 0.6 (Negation is multiplication by -1)**.**

$$-a = (-1) * a$$

Powers, square roots, and absolute values

Definition 0.4 (Power, base, and exponent)**.** Let a be a real number and n a natural number. The nth **power** of a is the product of n copies of a:

$$a^n \stackrel{\text{def}}{=} \underbrace{a \cdot a \cdot \cdots \cdot a}_{n \text{ factors}}$$

In a^n, a is the **base** and n is the **exponent**. The second power a^2 is the **square** of a.

Example 0.4 (Powers)**.**

- $$\begin{aligned} 2^3 &= 2 \cdot 2 \cdot 2 \\ &= 8, \end{aligned}$$

  with base 2 and exponent 3.

- $$\begin{aligned} (-3)^2 &= (-3) \cdot (-3) \\ &= 9, \end{aligned}$$

  but

  $$\begin{aligned} -3^2 &= -(3^2) \\ &= -9 : \end{aligned}$$

  the exponent applies only to the 3.

- $5^0 = 1$.

- $$\begin{aligned} 2^{-2} &= \frac{1}{2^2} \\ &= \frac{1}{4}. \end{aligned}$$

Definition 0.5 (Powers with exponent zero or a negative integer)**.** For the exponent 0 and negative integer exponents:

- $a^0 \stackrel{\text{def}}{=} 1$, including $0^0 = 1$ by convention;
- $a^{-n} \stackrel{\text{def}}{=} \frac{1}{a^n}$, when $a \neq 0$.

Remark (Powers with real exponents)

Remark 0.1 (Powers with real exponents). Definition 0.50 extends powers to real exponents when the base is positive.

Definition 0.6 (Square root)**.** For a real number $a \geq 0$, the **square root** of a, written $\sqrt{a}$, is the non-negative real number whose square (Definition 0.4) is a.

Example 0.5 (Square roots)**.**

- $\sqrt{9} = 3$. Although $(-3)^2 = 9$ too, -3 is negative, so $\sqrt{9}$ is not -3.
- $\sqrt{0} = 0$.
- $\sqrt{2} \approx 1.41421$.
- $\sqrt{-4}$ is not a real number, because no real number squares to -4 (see Section).

Definition 0.7 (Absolute value of a real number)**.** The **absolute value** of a real number a is

$$|a| \stackrel{\text{def}}{=} \begin{cases} a, & a \geq 0 \\ -a, & a < 0 \end{cases}$$

Example 0.6 (Absolute values)**.**

- $|2.5| = 2.5$, because $2.5 \geq 0$.
- $$\begin{aligned} |-3| &= -(-3) \\ &= 3, \end{aligned}$$

  because $-3 < 0$.
- $|0| = 0$.
- $$\begin{aligned} |-3| &= 3 \\ &= \sqrt{9} \\ &= \sqrt{(-3)^2} : \end{aligned}$$

  for every real a, $|a| = \sqrt{a^2}$ (Definition 0.6).
- $|x - 1| < 1$ says that x is less than 1 away from 1, that is, $0 < x < 2$.

Remark (Absolute value as distance)

Remark 0.2 (Absolute value as distance). For real numbers a and b, $|a - b|$ is the distance between a and b on the number line.

Minimum, maximum, argmin and argmax

Exercise 0.1 (Smallest value, and where it occurs)**.**

1. Let $f(x) = (x-2)^2 + 1$ for every $x \in \mathbb{R}$. What is the smallest value f takes, and at which x does it take that value?
2. Now let $g(x) = (x-2)^2 + 1$ only for $x \in \{0, 1, 3, 4\}$. What is the smallest value g takes, and at which x does it take that value?

Solution

Solution 0.1.

1. The square of a real number is never negative, so $(x-2)^2 \geq 0$ for every x, and therefore $f(x) \geq 0 + 1 = 1$. The square is 0 only when $x - 2 = 0$, that is, when $x = 2$, and there $f(2) = 1$. So the smallest value of f is 1, and $x = 2$ is the only input where f takes it.

2. Evaluate g at each of its four inputs:

x	0	1	3	4
$g(x)$	5	2	2	5

   The smallest value of g is 2, and g takes it at two inputs, $x = 1$ and $x = 3$. The answer to "where?" is the set $\{1, 3\}$, not a single number.

Definition 0.8 (Minimum)**.** Let $A \subseteq \mathbb{R}$. A number m is the **minimum** of A, written $\min A$, if $m \in A$ and $m \leq a$ for all $a \in A$.

Remark (Not every set has a minimum)

Remark 0.3 (Not every set has a minimum). The set $\{3, 1, 4\}$ has minimum 1, and the interval $[0, 1]$ has minimum 0. The interval $(0, 1]$ has no minimum: every element a of it has a smaller element, $a/2$, also in it.

Definition 0.9 (Maximum)**.** Let $A \subseteq \mathbb{R}$. A number M is the **maximum** of A, written $\max A$, if $M \in A$ and $a \leq M$ for all $a \in A$.

Example 0.7 (A maximum, and a set without one)**.**

- $\max\{-2, 5, 0\} = 5$: 5 is in the set, and $-2 \leq 5$, $5 \leq 5$, $0 \leq 5$.
- The interval $[0, 1)$ has no maximum. For any $a \in [0, 1)$, the number $\frac{a+1}{2}$ is also in $[0, 1)$, and it is larger than a, since $\frac{a+1}{2} > a$ exactly when $a < 1$. So no element of $[0, 1)$ is at least as large as every element.

Definition 0.10 (Argmin)**.** Let $f : A \to \mathbb{R}$ be a function. The **argmin** of f over A is the set of inputs where f takes its smallest value:

$$\arg\min_{x \in A} f(x) \stackrel{\text{def}}{=} \{x \in A : \forall a \in A,\ f(x) \leq f(a)\}$$

When this set has exactly one element $\hat{x}$, we write $\hat{x} = \arg\min_{x \in A} f(x)$.

Remark (Smallest value versus where it occurs)

Remark 0.4 (Smallest value versus where it occurs). The smallest value itself is $\min f(A)$, the minimum (Definition 0.8) of the image of f, and the argmin is where that value is attained. For example, let $f(x) = x^2$ on $A = \{-1, 0, 2\}$. The image is $f(A) = \{1, 0, 4\}$, so the smallest value is $\min f(A) = 0$, and $\arg\min_{x \in A} f(x) = \{0\}$.
The argmin is empty when f has no smallest value, for example $f(x) = x$ on $(0, 1]$.

Definition 0.11 (Argmax)**.** Let $f : A \to \mathbb{R}$ be a function. The **argmax** of f over A is the set of inputs where f takes its largest value:

$$\arg\max_{x \in A} f(x) \stackrel{\text{def}}{=} \{x \in A : \forall a \in A,\ f(a) \leq f(x)\}$$

When this set has exactly one element $\hat{x}$, we write $\hat{x} = \arg\max_{x \in A} f(x)$.

Example 0.8 (An argmax with two points, and an empty one)**.**

- Let $g(x) = x^2$ on $A = \{-2, 0, 1, 2\}$. The values are $g(-2) = 4$, $g(0) = 0$, $g(1) = 1$ and $g(2) = 4$, so the largest value is 4 and $\arg\max_{x \in A} g(x) = \{-2, 2\}$. This argmax has two elements, so it is not written as a single $\hat{x}$.
- Let $f(x) = x$ on $[0, 1)$. As in Example 0.7, for every input x the input $\frac{x+1}{2}$ has a larger value, so no input attains a largest value, and $\arg\max_{x \in [0,1)} f(x)$ is the empty set.

Global and local minimizers

Exercise 0.2 (A dip that is not the bottom)**.** Let $f(x) = x^3 - 3x$ for $x \in \mathbb{R}$.

1. Factor $f(x) - f(1)$ completely.
2. Use part 1 to compare $f(x)$ with $f(1)$ for every x in the interval $(0, 2)$. Is $f(x)$ larger than, smaller than, or equal to $f(1)$, and for which x?
3. Compute $f(-3)$. Is $f(1)$ the smallest value f takes on all of $\mathbb{R}$?
4. Does f take a smallest value anywhere on $\mathbb{R}$?

Solution

Solution 0.2.

1. Since

$$\begin{aligned} f(1) &= 1 - 3 \\ &= -2, \end{aligned}$$

so

$$\begin{aligned} f(x) - f(1) &= x^3 - 3x + 2 \\ &= (x-1)(x^2 + x - 2) \\ &= (x-1)(x-1)(x+2) \\ &= (x-1)^2(x+2). \end{aligned}$$

2. For $x \in (0, 2)$, $(x-1)^2 \geq 0$ and $x + 2 > 0$, so their product is at least 0. By part 1, $f(x) - f(1) \geq 0$, that is, $f(x) \geq f(1)$. So $f(x)$ is never smaller than $f(1)$ on $(0, 2)$. It equals $f(1)$ only when $(x-1)^2 = 0$, that is, only at $x = 1$, and it is larger than $f(1)$ at every other $x \in (0, 2)$.

3.
$$\begin{aligned} f(-3) &= -27 + 9 \\ &= -18. \end{aligned}$$

Since $-18 < -2 = f(1)$, $f(1)$ is not the smallest value of f on $\mathbb{R}$.

4. No. For each $n \geq 2$,

$$\begin{aligned} f(-n) &= -n^3 + 3n \\ &= -n(n^2 - 3) \\ &\leq -n, \end{aligned}$$

so no value of f is smaller than all the others: for any x, picking $n \geq 2$ with $-n < f(x)$ gives $f(-n) < f(x)$.

Definition 0.12 (Global minimizer)**.** Let $A \subseteq \mathbb{R}^p$ and let $f : A \to \mathbb{R}$ be a function. A point $x^* \in A$ is a **global minimizer** of f over A if $f(x^*) \leq f(x)$ for all $x \in A$.

Remark (Global minimizers form the argmin)

Remark 0.5 (Global minimizers form the argmin). The global minimizers of f are exactly the elements of $\arg\min_{x \in A} f(x)$ (Definition 0.10). A function can have more than one global minimizer. For example, $f(x) = (x^2 - 1)^2$ on $\mathbb{R}$ satisfies $f(x) \geq 0$ for every x, and $f(x) = 0$ exactly when $x = -1$ or $x = 1$, so its global minimizers are -1 and 1, and $\arg\min_{x \in \mathbb{R}} f(x) = \{-1, 1\}$.

Definition 0.13 (Local minimizer)**.** Let $A \subseteq \mathbb{R}^p$ and let $f : A \to \mathbb{R}$ be a function. A point $x^* \in A$ is a **local minimizer** of f if there is a number $\delta > 0$ such that $f(x^*) \leq f(x)$ for all $x \in A$ with $\|x - x^*\| < \delta$, where $\|\cdot\|$ is the Euclidean norm.

Remark (Local and global minimizers)

Remark 0.6 (Local and global minimizers). For $p = 1$, $\|x - x^*\| = |x - x^*|$. For example, with $x^* = 1$ and $\delta = 1$, the condition $|x - 1| < 1$ means $0 < x < 2$.
Every global minimizer (Definition 0.12) is a local minimizer: take any $\delta > 0$. For example, $x^* = 0$ is a global minimizer of $f(x) = x^2$ on $\mathbb{R}$, so it is also a local minimizer.
The converse fails. In Exercise 0.2, $x^* = 1$ is a local minimizer of $f(x) = x^3 - 3x$ (take $\delta = 1$), but not a global one.

Example 0.9 (A point that is not a local minimizer)**.** For $f(x) = x^3 - 3x$, the point $x = -1$ is not a local minimizer. For any t,

$$\begin{aligned}
f(-1+t) &= (-1+t)^3 - 3(-1+t) && \text{(substitute)} \\
&= (-1+3t-3t^2+t^3) - 3(-1+t) && \text{(expand the cube)} \\
&= -1+3t-3t^2+t^3+3-3t && \text{(multiply } -3 \text{ into the parentheses)} \\
&= 2-3t^2+t^3 && (3t-3t=0 \text{ and } -1+3=2) \\
&= 2+t^2(t-3) && \text{(factor out } t^2) \\
&= f(-1)+t^2(t-3), && (f(-1)=(-1)^3-3(-1), \text{ which equals } 2)
\end{aligned}$$

and $t^2(t-3) < 0$ whenever $t \neq 0$ and $t < 3$. So whatever $\delta > 0$ is, the point $x = -1 + t$ with $t = \min\{\delta/2, 1\}$ satisfies $|x-(-1)| < \delta$ and $f(x) < f(-1)$. For example,

$$\begin{aligned}
f(-0.9) &= -0.729 + 2.7 \\
&= 1.971 \\
&< 2.
\end{aligned}$$

Convex functions

Exercise 0.3 (A value compared with an average of values)**.** Let $f(x) = (x-2)^2$, and take the points $x = 0$ and $y = 4$ with $t = \frac{1}{2}$.

1. Compute $f(tx + (1-t)y)$.
2. Compute $tf(x) + (1-t)f(y)$.
3. Is $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$?

Solution

Solution 0.3.

1. With $t = \frac{1}{2}$,

$$\begin{aligned}
tx + (1-t)y &= \tfrac{1}{2} \cdot 0 + \tfrac{1}{2} \cdot 4 \\
&= 2,
\end{aligned}$$

so

$$f(2) = (2-2)^2$$
$$= 0.$$

2.
$$f(0) = (0-2)^2$$
$$= 4$$

and

$$f(4) = (4-2)^2$$
$$= 4,$$

so $\frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 4 = 4$.

3. Yes: $0 \leq 4$. The value of f at the midpoint of 0 and 4 is below the average of its values at the two endpoints.

Definition 0.14 (Line segment)**.** Let $x, y \in \mathbb{R}^p$. The **line segment** from x to y is the set of points

$$\{tx + (1-t)y : t \in [0,1]\}.$$

The value $t = 1$ gives x, the value $t = 0$ gives y, and $t = \frac{1}{2}$ gives the midpoint $\frac{1}{2}(x+y)$.

Example 0.10 (A line segment in $\mathbb{R}$)**.** For $p = 1$, $x = 0$ and $y = 4$, the line segment from x to y is the interval $[0, 4]$. The value $t = \frac{1}{4}$ gives the point $\frac{1}{4} \cdot 0 + \frac{3}{4} \cdot 4 = 3$.

Definition 0.15 (Chord)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ be a function, and let $x, y \in \mathbb{R}^p$. The **chord** of f from x to y is the line segment (Definition 0.14) in $\mathbb{R}^{p+1}$ from the point $(x, f(x))$ to the point $(y, f(y))$ of the graph of f:

$$\{(tx + (1-t)y,\ tf(x) + (1-t)f(y)) : t \in [0,1]\}.$$

Above the point $tx + (1-t)y$, the chord has height $tf(x) + (1-t)f(y)$.

Example 0.11 (A chord of x^2)**.** For $f(x) = x^2$, $x = -1$ and $y = 3$, the chord runs from $(-1, f(-1)) = (-1, 1)$ to $(3, f(3)) = (3, 9)$. With $t = \frac{1}{2}$, it passes through $\left(\frac{1}{2} \cdot (-1) + \frac{1}{2} \cdot 3,\ \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 9\right) = (1, 5)$, so above the point 1 it has height 5.

Definition 0.16 (Convex function)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ be a function. f is **convex** if

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

for all $x, y \in \mathbb{R}^p$ and all $t \in [0, 1]$.

Remark (Chords lie on or above the graph)

Remark 0.7 (Chords lie on or above the graph). The point $tx + (1-t)y$ lies on the line segment from x to y, and the right-hand side is the height above that point of the chord of f from x to y. So f is convex when every chord lies on or above the graph.
For example, take $f(x) = x^2$, $x = -1$, $y = 3$, and $t = \frac{1}{2}$. The point is $\frac{1}{2} \cdot (-1) + \frac{1}{2} \cdot 3 = 1$, where the graph has height $f(1) = 1$ and the chord has height

$$\begin{aligned} \tfrac{1}{2}f(-1) + \tfrac{1}{2}f(3) &= \tfrac{1}{2} \cdot 1 + \tfrac{1}{2} \cdot 9 \\ &= 5. \end{aligned}$$

The chord is above the graph: $1 \leq 5$.

Example 0.12 (A function that is not convex)**.** The function in Exercise 0.2, $f(x) = x^3 - 3x$, is not convex. Take $x = -2$, $y = 0$, $t = \frac{1}{2}$. The point is $\frac{1}{2} \cdot (-2) + \frac{1}{2} \cdot 0 = -1$, where the graph has height

$$\begin{aligned} f(-1) &= -1 + 3 \\ &= 2, \end{aligned}$$

but the chord has height

$$\begin{aligned} \tfrac{1}{2}f(-2) + \tfrac{1}{2}f(0) &= \tfrac{1}{2}(-8 + 6) + \tfrac{1}{2} \cdot 0 \\ &= -1, \end{aligned}$$

and $2 \leq -1$ is false.

Example 0.13 (A convex and a non-convex function)**.** $f(x) = (x-2)^2$ is convex (Definition 0.16): by expanding the square, $f(tx + (1-t)y) - tf(x) - (1-t)f(y) = -t(1-t)(x-y)^2 \leq 0$. Its local minimizer $x^* = 2$ is also a global minimizer, since

$$\begin{aligned} f(x) &= (x-2)^2 \\ &\geq 0 \\ &= f(2) \end{aligned}$$

for every x.

Without convexity, a local minimizer need not be global: $f(x) = x^3 - 3x$ is not convex (Example 0.12), and it has a local minimizer at $x^* = 1$ that is not global (Exercise 0.2).

Theorem 0.7 (Local minimizers of convex functions are global)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ be a convex function. Every local minimizer of f is a global minimizer of f over $\mathbb{R}^p$.*

Proof

Proof. Let x^* be a local minimizer of f, so there is a $\delta > 0$ with $f(x^*) \leq f(x)$ whenever $\|x - x^*\| < \delta$ (Definition 0.13). Suppose x^* is not a global minimizer (Definition 0.12). Then some $y \in \mathbb{R}^p$ has $f(y) < f(x^*)$, and in particular $y \neq x^*$.
Let $t = \min\left\{\frac{1}{2}, \frac{\delta}{2\|y - x^*\|}\right\}$, so $t \in (0, 1)$, and let $z = ty + (1-t)x^*$. Then $z - x^* = t(y - x^*)$, so $\|z - x^*\| = t\|y - x^*\| \leq \delta/2 < \delta$.
By convexity,

$$f(z) \leq tf(y) + (1-t)f(x^*) < tf(x^*) + (1-t)f(x^*) = f(x^*),$$

where the strict inequality uses $t > 0$ and $f(y) < f(x^*)$. So $f(z) < f(x^*)$ with $\|z - x^*\| < \delta$, which contradicts x^* being a local minimizer. Hence x^* is a global minimizer. □

Theorem 0.8 (Jensen's inequality for a weighted average)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ be a convex function. Let $x_1, \ldots, x_n \in \mathbb{R}^p$, and let $w_1, \ldots, w_n \geq 0$ be weights with $\sum_{i=1}^n w_i = 1$. Then*

$$f\left(\sum_{i=1}^n w_i x_i\right) \leq \sum_{i=1}^n w_i f(x_i).$$

Proof

Proof. The proof is by induction on n.
For $n = 1$ the weight is $w_1 = 1$, and both sides equal $f(x_1)$.
Now let $n \geq 2$, and assume the inequality holds for $n - 1$ points. If $w_n = 1$, the other weights are all 0, and both sides equal $f(x_n)$. Otherwise let

$$\begin{aligned} s &= 1 - w_n \\ &= \sum_{i=1}^{n-1} w_i \\ &> 0 \end{aligned}$$

and

$$y = \sum_{i=1}^{n-1} \frac{w_i}{s} x_i.$$

The weights w_i/s are non-negative and sum to 1, so the induction assumption gives

$$f(y) \leq \sum_{i=1}^{n-1} \frac{w_i}{s} f(x_i).$$

Since $\sum_{i=1}^{n} w_i x_i = sy + (1-s)x_n$ and $s \in (0, 1]$, the definition of a convex function (Definition 0.16) gives

$$f\left(\sum_{i=1}^{n} w_i x_i\right) = f(sy + (1-s)x_n) \leq sf(y) + (1-s)f(x_n).$$

Substituting the bound on $f(y)$, and using $1 - s = w_n$,

$$sf(y) + (1-s)f(x_n) \leq \sum_{i=1}^{n-1} w_i f(x_i) + w_n f(x_n) = \sum_{i=1}^{n} w_i f(x_i).$$

□

Example 0.14 (The average of the squares is at least the square of the average)**.** Take $f(x) = x^2$. It is convex by Definition 0.16: for $t \in [0, 1]$,

$$tx^2 + (1-t)y^2 - (tx + (1-t)y)^2 = t(1-t)(x-y)^2 \geq 0.$$

Take the three points $x_1 = 1$, $x_2 = 2$, $x_3 = 6$, and the equal weights

$$\begin{aligned} w_1 &= w_2 \\ &= w_3 \\ &= \tfrac{1}{3}. \end{aligned}$$

The weighted average of the points is their ordinary average, $\frac{1}{3}(1 + 2 + 6) = 3$, so the left side of Theorem 0.8 is $f(3) = 9$. The right side is the average of the squares, $\frac{1}{3}(1 + 4 + 36) = \frac{41}{3}$. The inequality holds: $9 \leq \frac{41}{3}$.
The gap is $\frac{41}{3} - 9 = \frac{14}{3}$. That is the average squared distance of the points from their average, $\frac{1}{3}((1-3)^2 + (2-3)^2 + (6-3)^2) = \frac{14}{3}$, which statistics calls the variance of the three points (about their mean, defined later). For $f(x) = x^2$, Jensen's inequality says this average squared distance is never negative.

Remark (The inequality reverses for concave functions)

Remark 0.8 (The inequality reverses for concave functions). If f is concave, then $-f$ is convex, and applying Theorem 0.8 to $-f$ reverses the inequality: $f\left(\sum_i w_i x_i\right) \geq \sum_i w_i f(x_i)$. For $f(x) = -x^2$ and the points in Example 0.14, $f(3) = -9 \geq -\frac{41}{3}$.

Infimum and supremum

Definition 0.17 (Upper and lower bounds)**.** Let $A \subseteq \mathbb{R}$.

- A real number u is an **upper bound** for A if $a \leq u$ for all $a \in A$.
- A real number ℓ is a **lower bound** for A if $\ell \leq a$ for all $a \in A$.

A is **bounded above** if it has an upper bound, **bounded below** if it has a lower bound, and **bounded** if it is both bounded above and bounded below. A real-valued function is bounded above, bounded below, or bounded if its image is.

Example 0.15 (Bounded and unbounded sets)**.**

- For $A = (0, 1]$, 1, 2, and 100 are upper bounds, and 0 and -5 are lower bounds, so A is bounded. 0.5 is not an upper bound, because $0.6 \in A$ and $0.6 > 0.5$.
- The natural numbers $\mathbb{N} = \{1, 2, 3, ...\}$ are bounded below, by 1, but not bounded above: for any real number u, some natural number n satisfies $n > u$.

Theorem 0.9 (Completeness of the real numbers)**.** *Let $A \subseteq \mathbb{R}$ be nonempty.*

- *If A is bounded above, the set of upper bounds of A has a minimum.*
- *If A is bounded below, the set of lower bounds of A has a maximum.*

Proof

Proof. The first statement is the least-upper-bound property of $\mathbb{R}$ (Rudin 1976, Definition 1.8, p. 4, and Theorem 1.19, p. 8). For the second, let $-A = \{-a : a \in A\}$. Negating both sides of an inequality reverses it (Theorem 0.4), so ℓ is a lower bound of A exactly when $-\ell$ is an upper bound of $-A$. The set $-A$ is nonempty and bounded above, so by the first statement its upper bounds have a minimum u, and then $-u$ is the largest lower bound of A. □

Example 0.16 (Completeness for an interval)**.** For $A = (1, 2]$, the lower bounds of A are the numbers $\ell \leq 1$, and the largest of them is 1. The upper bounds of A are the numbers $u \geq 2$, and the smallest of them is 2.

Definition 0.18 (Infimum (greatest lower bound))**.** Let $A \subseteq \mathbb{R}$ be nonempty and bounded below (Definition 0.17). The **infimum** of A, written $\inf A$, is the greatest real number t satisfying $t \leq a$ for all $a \in A$:

$$\inf A \stackrel{\text{def}}{=} \max\{t \in \mathbb{R} : \forall a \in A,\ t \leq a\}$$

If A is nonempty but not bounded below, we write $\inf A = -\infty$ by convention.

Remark (Existence of the infimum, and when it is a minimum)

Remark 0.9 (Existence of the infimum, and when it is a minimum). The maximum in Definition 0.18 always exists, by the completeness of the real numbers (Theorem 0.9). For example, for $A = (1, 2]$, the numbers t with $t \leq a$ for all $a \in A$ are those with $t \leq 1$, and the largest of them is 1, so $\inf A = 1$.
If the infimum belongs to A, it equals the minimum: $\inf A = \min A$. For example,

$$\begin{aligned} \inf[1, 2] &= 1 \\ &= \min[1, 2]. \end{aligned}$$

For $A = (1, 2]$, the infimum 1 is not in A, and A has no minimum.

Example 0.17 (Numerical examples of infimum)**.**

- $\inf\{1, 2, 3\} = 1$, since 1 is the smallest element.
- $$\begin{aligned} \inf(0.5, 1] &= 0.5 \\ &= \min[0.5, 1] : \end{aligned}$$

  for intervals open below, the infimum equals the minimum of the corresponding closed-below interval, even though $0.5 \notin (0.5, 1]$. More generally,

  $$\begin{aligned} \inf(c, b] &= \min[c, b] \\ &= c \end{aligned}$$

  for any $c < b$.
- $\inf\{t \geq 0 : t > 0.5\} = 0.5$, even though 0.5 itself is not in the set.

- $\inf\{-1,-2,-3,\ldots\} = -\infty$, because no real number is less than or equal to every element of that set.

Definition 0.19 (Supremum (least upper bound))**.** Let $A \subseteq \mathbb{R}$ be nonempty and bounded above (Definition 0.17). The **supremum** of A, written $\sup A$, is the smallest real number t satisfying $a \leq t$ for all $a \in A$:

$$\sup A \stackrel{\text{def}}{=} \min\{t \in \mathbb{R} : \forall a \in A,\ a \leq t\}$$

If A is nonempty but not bounded above, we write $\sup A = +\infty$ by convention.

Remark (Existence of the supremum, and when it is a maximum)

Remark 0.10 (Existence of the supremum, and when it is a maximum). The minimum in Definition 0.19 always exists, by the completeness of the real numbers (Theorem 0.9). For example, for $A = [1,2)$, the numbers t with $a \leq t$ for all $a \in A$ are those with $t \geq 2$, and the smallest of them is 2, so $\sup A = 2$.
If the supremum belongs to A, it equals the maximum: $\sup A = \max A$. For example,

$$\begin{aligned}\sup[1,2] &= 2 \\ &= \max[1,2].\end{aligned}$$

For $A = [1,2)$, the supremum 2 is not in A, and A has no maximum.

Example 0.18 (Numerical examples of supremum)**.**

- $\sup\{1,2,3\} = 3$, since 3 is the largest element.
- $\sup\{t \geq 0 : t < 0.5\} = 0.5$, even though 0.5 itself is not in the set.
- $\sup\{1,2,3,\ldots\} = +\infty$, because no real number is greater than or equal to every element of that set.

Sums

Definition 0.20 (Term of a sum)**.** In a sum $a_1 + a_2 + \cdots + a_n$, each of the numbers or expressions $a_1, a_2, \ldots, a_n$ being added is a **term** of the sum.

Example 0.19 (Terms of a sum)**.**

- The sum $3 + 5 + 9$ has three terms: 3, 5 and 9.
- The sum $x^2 - 3x + 7 = x^2 + (-3x) + 7$ has three terms: x^2, $-3x$ and 7. A subtracted

expression counts as a term with a minus sign.

Definition 0.21 (Identity element)**.** Let S be a set, such as the real numbers $\mathbb{R}$, and let $\star$ be an operation that combines two elements a and b of S into an element $a \star b$ of S, such as addition $(a + b)$ or multiplication $(a \times b)$ of real numbers. An element u of S is an **identity element** for $\star$ if $a \star u = a$ and $u \star a = a$ for every $a \in S$.

Example 0.20 (Identity elements for addition and multiplication)**.**

- 0 is the identity element for addition: for example, $5 + 0 = 5$ and $0 + 5 = 5$ (Theorem 0.10).
- 1 is the identity element for multiplication: for example, $5 \times 1 = 5$ and $1 \times 5 = 5$ (Theorem 0.13).
- 0 is not an identity element for subtraction: $5 - 0 = 5$, but $0 - 5 = -5 \neq 5$.

Definition 0.22 (Commutative operation)**.** An operation $\star$ on a set S (Definition 0.21) is **commutative** if $a \star b = b \star a$ for all $a, b \in S$: the order of the two inputs does not matter. Some sources, including the titles of Theorem 0.11 and Theorem 0.14, call a commutative operation **symmetric**.

Example 0.21 (A commutative operation, and one that is not)**.**

- Addition is commutative: for example,

$$\begin{aligned} 2 + 5 &= 7 \\ &= 5 + 2. \end{aligned}$$

- Subtraction is not commutative: $5 - 3 = 2$, but $3 - 5 = -2$.

Definition 0.23 (Associative operation)**.** An operation $\star$ on a set S (Definition 0.21) is **associative** if $(a \star b) \star c = a \star (b \star c)$ for all $a, b, c \in S$: which pair is combined first does not matter.

Example 0.22 (An associative operation, and one that is not)**.**

- Multiplication is associative: for example,

$$(2 \times 3) \times 4 = 6 \times 4$$
$$= 24$$

and

$$2 \times (3 \times 4) = 2 \times 12$$
$$= 24.$$

- Subtraction is not associative:

$$(8 - 4) - 2 = 4 - 2$$
$$= 2,$$

but

$$8 - (4 - 2) = 8 - 2$$
$$= 6.$$

Theorem 0.10 (Adding zero changes nothing)**.**

$$a + 0 = a$$

Theorem 0.11 (Sums are symmetric)**.**

$$a + b = b + a$$

Theorem 0.12 (Sums are associative)**.** *When adding three numbers, it does not matter which pair you add first:*

$$(a + b) + c = a + (b + c)$$

Example 0.23 (Grouping a sum two ways)**.**

$$(2 + 3) + 4 = 5 + 4$$
$$= 9,$$

and

$$2 + (3 + 4) = 2 + 7$$
$$= 9.$$

Products

Theorem 0.13 (Multiplying by 1 changes nothing)**.**

$$a \times 1 = a$$

Theorem 0.14 (Products are symmetric)**.**

$$a \times b = b \times a$$

Theorem 0.15 (Products are associative)**.**

$$(a \times b) \times c = a \times (b \times c)$$

Division

Theorem 0.16 (Division can be written as a product)**.** *If $b \neq 0$, then*

$$\frac{a}{b} = a \times \frac{1}{b}$$

Sums and products together

Definition 0.24 (Distributes over)**.** An operation $\star$ on a set S **distributes over** an operation $\diamond$ on S if

$$a \star (b \diamond c) = (a \star b) \diamond (a \star c)$$

for all $a, b, c \in S$.

Example 0.24 (Addition does not distribute over multiplication)**.**

$$\begin{aligned} 2 + (3 \times 4) &= 2 + 12 \\ &= 14, \end{aligned}$$

but

$$\begin{aligned} (2 + 3) \times (2 + 4) &= 5 \times 6 \\ &= 30. \end{aligned}$$

So Definition 0.24 fails with $\star = +$ and $\diamond = \times$.

Definition 0.25 (Distributive law)**.** The **distributive law** is the statement that multiplication distributes over addition (Definition 0.24): $a \times (b + c) = (a \times b) + (a \times c)$ (Theorem 0.17). We then say multiplication is **distributive**.

Example 0.25 (Multiplication distributes over addition)**.**

$$\begin{aligned} 3 \times (4 + 5) &= 3 \times 9 \\ &= 27, \end{aligned}$$

and

$$\begin{aligned} (3 \times 4) + (3 \times 5) &= 12 + 15 \\ &= 27. \end{aligned}$$

Theorem 0.17 (Multiplication is distributive)**.**

$$a(b + c) = ab + ac$$

Definition 0.26 (Like terms)**.** Two terms of a sum are **like terms** if they are the same product of variables, each to the same power, possibly multiplied by different constants: $c_1 m$ and $c_2 m$, where c_1 and c_2 are constants and m is that product. To **collect** like terms is to replace their sum by one term, using the distributive law (Theorem 0.17): $c_1 m + c_2 m = (c_1 + c_2)m$.

Example 0.26 (Like and unlike terms)**.**

- $3ab$ and $-5ab$ are like terms; collecting them gives

$$\begin{aligned} 3ab + (-5ab) &= (3 - 5)ab \\ &= -2ab. \end{aligned}$$

- $2x$ and $2x^2$ are not like terms: x appears to different powers.

Exercise 0.4 (Expand a squared sum)**.** Is $(3+4)^2$ equal to 3^2+4^2? Then expand $(a+b)^2$ for any numbers a and b, using only the distributive law and the rules above.

Solution

Solution 0.4. No:

$$\begin{aligned}(3+4)^2 &= 7^2 \\ &= 49,\end{aligned}$$

while

$$\begin{aligned}3^2 + 4^2 &= 9 + 16 \\ &= 25.\end{aligned}$$

The difference, $49 - 25 = 24$, is $2 \cdot 3 \cdot 4$.
To see where that extra term comes from, write the square as a product and apply the distributive law (Theorem 0.17) twice:

$$\begin{aligned}(a+b)^2 &= (a+b)(a+b) && \text{(definition of a square)} \\ &= (a+b)\,a + (a+b)\,b && \text{(distributive law)} \\ &= (a^2 + ba) + (ab + b^2) && \text{(distributive law, twice)} \\ &= a^2 + ab + ab + b^2 && \text{(commutative and associative laws)} \\ &= a^2 + 2ab + b^2 && \text{(collect like terms, Definition 0.26)}\end{aligned}$$

The step "commutative and associative laws" uses Theorem 0.14 to write ba as ab, and Theorem 0.12 to drop the parentheses.

Theorem 0.18 (Square of a sum)**.** *For any numbers a and b,*

$$(a+b)^2 = a^2 + 2ab + b^2$$

Proof

Proof. By Solution 0.4. □

Remark (Square of a difference)

Remark 0.11 (Square of a difference). Replacing b by $-b$ in Theorem 0.18 gives $(a-b)^2 = a^2 - 2ab + b^2$. For example, the square $(y - \hat{y})^2$ of the difference between an observed value y and a prediction $\hat{y}$ of it expands this way. With $y = 5$ and $\hat{y} = 3$,

$$\begin{aligned}(5-3)^2 &= 2^2 \\ &= 4,\end{aligned}$$

and

$$5^2 - 2 \cdot 5 \cdot 3 + 3^2 = 25 - 30 + 9$$
$$= 4.$$

Summation notation

Exercise 0.5 (Expand a sum)**.** The expression $\sum_{i=1}^{4} i^2$ is shorthand for a sum of four terms.

1. Guess which four terms, and add them up.
2. Write $\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$ for $N = 3$ without the $\sum$ symbol.

Solution

Solution 0.5.

1. Replace i by each of $1, 2, 3, 4$ in turn, and add the results:

$$\sum_{i=1}^{4} i^2 = 1^2 + 2^2 + 3^2 + 4^2$$
$$= 1 + 4 + 9 + 16$$
$$= 30$$

2. Replace i by each of $1, 2, 3$ in turn inside the parentheses, add the three squares, and multiply the total by $\frac{1}{3}$:

$$\frac{1}{3}\sum_{i=1}^{3}(y_i - \hat{y}_i)^2 = \frac{1}{3}[(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2]$$

Definition 0.27 (Summation notation)**.** Let m and n be integers with $m \leq n$, and let $a_m, a_{m+1}, \ldots, a_n$ be numbers. The **sum** of a_m through a_n is

$$\sum_{i=m}^{n} a_i \stackrel{\text{def}}{=} a_m + a_{m+1} + \cdots + a_n$$

The variable i is the **index** of the sum; m and n are its **lower** and **upper limits**.

Remark (The index is a placeholder)

Remark 0.12 (The index is a placeholder). The name of the index does not change the sum: $\sum_{i=1}^{n} a_i$ and $\sum_{j=1}^{n} a_j$ are the same number. For example,

$$\begin{aligned}\sum_{i=1}^{3} i &= 1+2+3 \\ &= 6\end{aligned}$$

and

$$\begin{aligned}\sum_{j=1}^{3} j &= 1+2+3 \\ &= 6.\end{aligned}$$

Definition 0.28 (Empty sum)**.** When the upper limit is less than the lower limit ($n < m$), the sum $\sum_{i=m}^{n} a_i$ has no terms. By convention, such an **empty sum** equals 0. For example, $\sum_{i=1}^{0} a_i = 0$.

Exercise 0.6 (Rearrange a sum)**.** Let c be a number, and let a_1, a_2, a_3 and b_1, b_2, b_3 be numbers. Using Definition 0.27 and the rules of algebra above, show that:

1. $\sum_{i=1}^{3} c\,a_i = c\sum_{i=1}^{3} a_i$;
2. $\sum_{i=1}^{3}(a_i + b_i) = \sum_{i=1}^{3} a_i + \sum_{i=1}^{3} b_i$.

Does either argument depend on there being exactly three terms?

Solution

Solution 0.6. Each step below applies Definition 0.27, the distributive law (Theorem 0.17), or the commutative and associative laws of addition (Theorem 0.11 and Theorem 0.12).

1. Expand the sum, then factor out c:

$$\begin{aligned}\sum_{i=1}^{3} c\,a_i &= c\,a_1 + c\,a_2 + c\,a_3 && \text{(expand the sum)} \\ &= c(a_1 + a_2 + a_3) && \text{(distributive law)} \\ &= c\sum_{i=1}^{3} a_i && \text{(collect the sum)}\end{aligned}$$

2. Expand the sum, then regroup the terms:

$$\begin{aligned}\sum_{i=1}^{3}(a_i+b_i) &= (a_1+b_1)+(a_2+b_2)+(a_3+b_3) && \text{(expand the sum)}\\ &= (a_1+a_2+a_3)+(b_1+b_2+b_3) && \text{(commutative and associative laws)}\\ &= \sum_{i=1}^{3} a_i + \sum_{i=1}^{3} b_i && \text{(collect the sum)}\end{aligned}$$

Neither argument uses the number of terms: the same steps work for any lower and upper limits.

Theorem 0.19 (A constant factor comes out of a sum)**.** *For any number c and numbers $a_m, \dots, a_n$,*

$$\sum_{i=m}^{n} c\, a_i = c \sum_{i=m}^{n} a_i$$

Proof

Proof. By Solution 0.6, part 1. □

Theorem 0.20 (A sum of sums splits)**.** *For any numbers $a_m, \dots, a_n$ and $b_m, \dots, b_n$,*

$$\sum_{i=m}^{n}(a_i+b_i) = \sum_{i=m}^{n} a_i + \sum_{i=m}^{n} b_i$$

Proof

Proof. By Solution 0.6, part 2. □

Exercise 0.7 (Sum over a set)**.** Let $A = \{-1, 2, 5\}$ and let $f(x) = x^2$. The expression $\sum_{x\in A} f(x)$ means: add up $f(x)$ for each element x of A.

1. Write $\sum_{x\in A} f(x)$ without the $\sum$ symbol, and compute its value.
2. List the elements of A in a different order, and add up the same terms in that order. Is the total the same?

Solution

Solution 0.7.

1. Replace x by each element of A in turn, and add the results:

$$\begin{aligned}\sum_{x\in A} f(x) &= f(-1) + f(2) + f(5)\\ &= (-1)^2 + 2^2 + 5^2\\ &= 1 + 4 + 25\\ &= 30\end{aligned}$$

2. In the order $5, -1, 2$, the terms are 25, 1 and 4. Their total is $25 + 1 + 4 = 30$. The total is the same, because addition does not depend on the order of the terms.

Definition 0.29 (Sum over a finite set)**.** Let $A = \{x_1, x_2, \dots, x_k\}$ be a finite set with $k \geq 1$ different elements. Let f be a function that gives a number $f(x)$ for each element x of A. The **sum of f over A** is

$$\sum_{x\in A} f(x) \stackrel{\text{def}}{=} f(x_1) + f(x_2) + \cdots + f(x_k)$$

If A has no elements, the sum has no terms, and by convention it equals 0.

Remark (The order of the terms does not matter)

Remark 0.13 (The order of the terms does not matter). The order in which we list the elements of A does not change the sum. The reason is that addition is commutative and associative: we can reorder and regroup the terms of a finite sum without changing the total. For example, if $A = \{1, 2, 3\}$ and $f(x) = x^2$, listing A as $1, 2, 3$ gives $1 + 4 + 9 = 14$, and listing A as $3, 1, 2$ gives $9 + 1 + 4 = 14$. Exercise 0.7, part 2, shows another example. When $A = \{m, m + 1, \dots, n\}$, this sum is the same as $\sum_{i=m}^{n} f(i)$ from Definition 0.27.

Remark (Leaving the set out)

Remark 0.14 (Leaving the set out). Some authors leave the set out and write $\sum_x f(x)$. This shorthand means the sum over every value x can take, $\sum_{x\in\mathcal{R}(x)} f(x)$, where $\mathcal{R}(x)$ is the range of x. For example, if x is the outcome of one roll of a six-sided die, $\sum_x f(x)$ means $\sum_{x\in\{1,2,3,4,5,6\}} f(x)$. In these notes, we write the set out in full (see Notational shorthands).

Example 0.27 (A sum over the empty set)**.** For example, $\sum_{x \in \{\}} x^2 = 0$.

Quotients

Definition 0.30 (Quotient)**.** For real numbers a and b with $b \neq 0$, the **quotient** of a by b is the result of dividing a by b:

$$\frac{a}{b}$$

Definition 0.31 (Fraction, numerator, and denominator)**.** A quotient $\frac{a}{b}$ (Definition 0.30) is also called a **fraction**; a is its **numerator** and b its **denominator**. For example, the fraction $\frac{6}{4}$ has numerator 6 and denominator 4.

Example 0.28 (A quotient)**.** The quotient of 6 by 4 is $\frac{6}{4} = 1.5$. The quotient of 6 by 0 is undefined, because Definition 0.30 requires a nonzero denominator.

Definition 0.32 (Follow-up and person-years)**.** In a study that observes participants over time, a participant's **follow-up** is the length of time the study observes that participant, and the study's **person-years** are the sum of the follow-up times of all participants, measured in years.

Example 0.29 (Counting person-years)**.** If 10 people are each followed for 2 years, the study has $10 \cdot 2 = 20$ person-years. If one person is followed for 3 years and another for $\frac{1}{2}$ year, the study has $3 + \frac{1}{2} = 3.5$ person-years.

Definition 0.33 (Rate)**.** A **rate** is a quotient of two quantities, usually with a denominator that measures time, such as weeks or person-years of follow-up (Definition 0.32). For example, 12 new cases in 4 weeks is a rate of $\frac{12}{4} = 3$ new cases per week, and 30 cases over 10,000 person-years of follow-up is a rate of $\frac{30}{10{,}000} = 0.003$ cases per person-year.

cf. https://en.wikipedia.org/wiki/Rate_(mathematics)

Definition 0.34 (Ratios)**.** A **ratio** is a quotient in which the numerator and denominator are measured using the same unit scales.
cf. https://en.wikipedia.org/wiki/Ratio

Example 0.30 (A ratio, and a quotient that is not one)**.**

- A board 150 cm long and one 75 cm long have length ratio $\frac{150\text{ cm}}{75\text{ cm}} = 2$: both lengths are in centimeters, so the units cancel and the ratio has none.
- A sample of mass 300 g and volume 150 cm^3 gives the quotient $\frac{300\text{ g}}{150\text{ cm}^3} = 2$ g per cm^3, its density. The numerator and denominator are in different units, so this quotient is not a ratio.

Definition 0.35 (Proportion)**.** In statistics, a **proportion** typically means a ratio where the numerator represents a subset of the denominator.
See https://en.wikipedia.org/wiki/Population_proportion.
See also https://en.wikipedia.org/wiki/Proportion_(mathematics) for other meanings.

Example 0.31 (A proportion, and a ratio that is not one)**.** In a clinic with 120 patients, 30 of whom smoke:

- the proportion of patients who smoke is $\frac{30}{120} = 0.25$: the 30 smokers are a subset of the 120 patients;
- the ratio of smokers to non-smokers is $\frac{30}{90} = \frac{1}{3}$. Both counts are of patients, so this quotient is a ratio, but it is not a proportion: the 30 smokers are not part of the 90 non-smokers.

Definition 0.36 (Proportional)**.** Two functions $f(x)$ and $g(x)$ are **proportional** if their ratio $\frac{f(x)}{g(x)}$ does not depend on x. (cf. https://en.wikipedia.org/wiki/Proportionality_(mathematics))

Example 0.32 (Proportional and non-proportional functions)**.**

- $f(x) = 6x^2$ and $g(x) = 2x^2$ are proportional: for $x \neq 0$,

$$\begin{aligned}\frac{f(x)}{g(x)} &= \frac{6x^2}{2x^2} \\ &= 3,\end{aligned}$$

  which does not depend on x.

- $f(x) = x + 1$ and $g(x) = x$ are not proportional: for $x \neq 0$,

$$\begin{aligned}\frac{f(x)}{g(x)} &= \frac{x+1}{x} \\ &= \frac{x}{x} + \frac{1}{x} \\ &= 1 + \frac{1}{x},\end{aligned}$$

which is 2 at $x = 1$ and $\frac{3}{2}$ at $x = 2$.

Additional reference for elementary algebra: https://en.wikipedia.org/wiki/Population_proportion#Mathematical_definition

Polynomials

Exercise 0.8 (Reading a polynomial)**.** Let $f(x) = 3x^4 - x + 7$.

1. Write f in the form $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, listing every coefficient a_4, a_3, a_2, a_1, a_0.
2. What is the degree of f, and what is its leading coefficient?

Solution

Solution 0.8.

1. $f(x) = 3x^4 + 0 \cdot x^3 + 0 \cdot x^2 + (-1) \cdot x + 7$, so $a_4 = 3$, $a_3 = 0$, $a_2 = 0$, $a_1 = -1$, and $a_0 = 7$. Powers of x that do not appear have coefficient 0.
2. The highest power with a nonzero coefficient is x^4, so the degree is 4 and the leading coefficient is $a_4 = 3$.

Definition 0.37 (Constant function)**.** A function $f : A \to B$ is **constant** if there is some $c \in B$ with $f(x) = c$ for all $x \in A$.

Example 0.33 (Constant and non-constant functions)**.**

- $f(x) = 7$ on $\mathbb{R}$ is constant, with $c = 7$.
- $f(x) = x^2$ on $\mathbb{R}$ is not constant: $f(0) = 0$ but $f(1) = 1$.

Definition 0.38 (Polynomial)**.** A **polynomial** (in one variable x) is a function $f : \mathbb{R} \to \mathbb{R}$

that can be written as

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

for some integer $n \geq 0$ and constants $a_0, a_1, \ldots, a_n \in \mathbb{R}$ with $a_n \neq 0$. The constants $a_0, a_1, \ldots, a_n$ are the **coefficients** of f, with a_k the coefficient of x^k, and the terms of f are $a_n x^n, \ldots, a_1 x, a_0$.

Remark (Constant and zero polynomials)

Remark 0.15 (Constant and zero polynomials). A constant function $f(x) = 7$ is a polynomial with $n = 0$ and $a_0 = 7$. The requirement $a_n \neq 0$ means this definition covers nonzero polynomials only: the zero function $f(x) = 0$ is excluded here, because it has no nonzero coefficient to serve as a_n.

Definition 0.39 (Degree of a polynomial)**.** Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with $a_n \neq 0$. The integer n is the **degree** of f.

Example 0.34 (Degrees of some polynomials)**.**

- $f(x) = 4x$ has degree 1.
- $f(x) = 5 - 2x^2 + x^3$ has degree 3: the degree is the highest power with a nonzero coefficient, not the power in the first term written.
- A constant polynomial $f(x) = 7$ has degree 0.

Definition 0.40 (Leading coefficient)**.** Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial of degree n. The constant a_n is the **leading coefficient** of f.

Example 0.35 (Leading coefficients)**.**

- $f(x) = 5 - 2x^2 + x^3$ has degree 3, so its leading coefficient is $a_3 = 1$, not the 5 written first.
- $f(x) = 3 - x^2$ has leading coefficient $a_2 = -1$.

Definition 0.41 (Quadratic and cubic polynomials)**.** A polynomial of degree 2, $f(x) = a_2 x^2 + a_1 x + a_0$ with $a_2 \neq 0$, is a **quadratic** polynomial. A polynomial of degree 3 is a **cubic** polynomial.

Example 0.36 (Quadratic and cubic polynomials)**.**

- $$\begin{aligned} f(x) &= (x-2)^2 \\ &= x^2 - 4x + 4 \end{aligned}$$

  is quadratic, by Remark 0.11 with $a = x$ and $b = 2$.
- $f(x) = x^3 - 3x$ is cubic.
- $f(x) = 4x + 1$ is neither: it has degree 1.

Definition 0.42 (Parabola)**.** A **parabola** is the graph of a quadratic polynomial (Definition 0.41).

Example 0.37 (The parabola $y = x^2$)**.** The graph of $f(x) = x^2$ is a parabola. It contains the points $(-1, 1)$, $(0, 0)$ and $(2, 4)$, and its lowest point is $(0, 0)$, since $x^2 \geq 0$ for every x.

Affine and linear functions of one variable

Definition 0.43 (Affine function of one variable)**.** A function $f : \mathbb{R} \to \mathbb{R}$ is **affine** if there are numbers m and b such that

$$f(x) = mx + b \quad \text{for all } x \in \mathbb{R}$$

In this formula, m is called the **slope** of f and b its **intercept**.

Definition 0.44 (Linear function of one variable)**.** A function $f : \mathbb{R} \to \mathbb{R}$ is **linear** if it is affine (Definition 0.43) with intercept 0; that is, if there is a number m such that

$$f(x) = mx \quad \text{for all } x \in \mathbb{R}$$

Theorem 0.21 (Intercept and slope of an affine function)**.** *Let $f(x) = mx + b$ be an affine function (Definition 0.43).*

1. *The intercept is the value at zero: $f(0) = b$.*
2. *The slope is the change in f per unit change in x: for any $x_1 \neq x_2$,*

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = m$$

3. *So f determines its slope and intercept: if also $f(x) = m'x + b'$ for all x, then $m' = m$ and $b' = b$.*

Proof

Proof. **Part 1.**

$$\begin{aligned} f(0) &= m \cdot 0 + b && \text{(Definition 0.43)} \\ &= 0 + b && \text{(any number times 0 is 0)} \\ &= b + 0 && \text{(Theorem 0.11)} \\ &= b && \text{(Theorem 0.10)} \end{aligned}$$

Part 2. First the numerator:

$$\begin{aligned} f(x_2) - f(x_1) &= (mx_2 + b) - (mx_1 + b) && \text{(Definition 0.43)} \\ &= mx_2 + b - mx_1 - b && \text{(subtracting a sum subtracts each term)} \\ &= mx_2 - mx_1 + b - b && \text{(Theorem 0.11, swapping } b \text{ and } -mx_1) \\ &= (mx_2 - mx_1) + (b - b) && \text{(Theorem 0.12)} \\ &= (mx_2 - mx_1) + 0 && (b - b = 0) \\ &= mx_2 - mx_1 && \text{(Theorem 0.10)} \\ &= mx_2 + m(-x_1) && (-(mx_1) = m(-x_1), \text{ by Theorem 0.6 and Theorem 0.15}) \\ &= m(x_2 + (-x_1)) && \text{(Theorem 0.17, read right to left)} \\ &= m(x_2 - x_1) && \text{(adding a negative is subtracting)} \end{aligned}$$

Then, writing $d = x_2 - x_1$, which is not 0 because $x_1 \neq x_2$,

$$\begin{aligned} \frac{f(x_2) - f(x_1)}{x_2 - x_1} &= \frac{md}{d} && \text{(the numerator above)} \\ &= (md) \cdot \frac{1}{d} && \text{(Theorem 0.16)} \\ &= m \cdot \left(d \cdot \frac{1}{d} \right) && \text{(Theorem 0.15)} \\ &= m \cdot \frac{d}{d} && \text{(Theorem 0.16)} \\ &= m \cdot 1 && \text{(a nonzero number divided by itself is 1)} \\ &= m && \text{(Theorem 0.13)} \end{aligned}$$

Part 3. Part 1, applied to each formula, gives

$$\begin{aligned} b &= f(0) \\ &= b'. \end{aligned}$$

Part 2 with $x_1 = 0$ and $x_2 = 1$, applied to each formula, gives

$$\begin{aligned} m &= f(1) - f(0) \\ &= m'. \end{aligned}$$

□

Example 0.38 (An affine function, and a linear one)**.** $f(x) = 2x + 3$ is affine with slope 2 and intercept 3. By Theorem 0.21, $f(0) = 3$, and each unit step in x raises f by 2: for example, $f(1) = 5$ and $f(2) = 7$. Its intercept is not 0, so it is not linear.
$g(x) = -\frac{1}{2}x$ is affine with intercept 0, so it is linear: $g(0) = 0$, and each unit step in x lowers g by $\frac{1}{2}$.

ℹ Remark (Elementary algebra calls $mx + b$ "linear")

Remark 0.16 (Elementary algebra calls $mx + b$ "linear"). Elementary algebra usually calls $f(x) = mx + b$ a *linear function*, because its graph is a straight line. This site calls that an affine function and keeps "linear" for the case $b = 0$, because that is the sense used in linear algebra, where a linear map must send 0 to 0 (theorem).

Limits of sequences

Definition 0.45 (Limit of a sequence)**.** A sequence (a_n) of real numbers **converges** to a real number L, written $\lim_{n\to\infty} a_n = L$ or $a_n \to L$, if for every $\varepsilon > 0$ there is a natural number N such that

$$|a_n - L| < \varepsilon \quad \text{for every } n \geq N.$$

Then L is the **limit** of the sequence. A sequence that converges to some real number is **convergent**, and a sequence that does not converge to any real number **diverges**.

Example 0.39 (A convergent sequence and a divergent one)**.**

- $a_n = \frac{1}{n}$ converges to 0. Given $\varepsilon > 0$, take N to be any natural number larger than $\frac{1}{\varepsilon}$. For every $n \geq N$, $\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon$. For example, with $\varepsilon = 0.01$, take $N = 101$: every $n \geq 101$ has $\frac{1}{n} \leq \frac{1}{101} < 0.01$.
- $c_n = n$ diverges. For any real number L and $\varepsilon = 1$, every $n > L + 1$ has $|n - L| > 1$, so no N works. For example, with $L = 5$, every $n \geq 7$ has $|n - 5| \geq 2$. It diverges to ∞: for every real number M, every $n \geq N$ has $c_n = n > M$ when N is a natural number larger than M.

Definition 0.46 (Divergence of a sequence to infinity)**.** A sequence (a_n) of real numbers **diverges to** ∞, written $\lim_{n\to\infty} a_n = \infty$, if for every real number M there is a natural number N such that $a_n > M$ for every $n \geq N$.

Exponentials and Logarithms

Definition 0.47 (Exponential function)**.** The **exponential function** $\exp : \mathbb{R} \to (0, \infty)$ is

$$\exp\{x\} \stackrel{\text{def}}{=} \lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n,$$

the limit of the sequence $(1 + \frac{x}{1})^1, (1 + \frac{x}{2})^2, (1 + \frac{x}{3})^3, \ldots$. That limit exists for every real number x, and it is positive.

Example 0.40 (Approximating $\exp\{1\}$)**.** For $x = 1$, the terms $\left(1 + \frac{1}{n}\right)^n$ are:

n	$\left(1 + \frac{1}{n}\right)^n$
1	2
10	2.59374...
100	2.70481...
1000	2.71692...

They approach $\exp\{1\} = 2.71828\ldots$. For $x = 0$, every term is $(1 + 0)^n = 1$, so $\exp\{0\} = 1$.

Definition 0.48 (Euler's number)**. Euler's number** is

$$e \stackrel{\text{def}}{=} \exp\{1\} = 2.71828\ldots$$

Example 0.41 (Euler's number is irrational)**.** $e = 2.71828\ldots$ is an irrational number, so no fraction equals it exactly. The fraction $\frac{19}{7} = 2.714\ldots$ is close, but $\frac{19}{7} \neq e$ (see Wikipedia: e (mathematical constant)).

Definition 0.49 (Natural logarithm)**.** For a real number $a > 0$, the **natural logarithm** of a, written $\log\{a\}$, is the real number y with $\exp\{y\} = a$. The exponential function takes each positive value at exactly one input, so there is exactly one such y, and $\log : (0, \infty) \to \mathbb{R}$ is the inverse function of exp (Definition 0.47).

Remark (Other notations for the natural logarithm)

Remark 0.17 (Other notations for the natural logarithm). In these notes, $\log\{x\}$ is always the natural logarithm, the logarithm with base e (Definition 0.48). Some sources write $\ln x$ for the natural logarithm and reserve $\log x$ for the logarithm with base 10.

Example 0.42 (Natural logarithms)**.**

- $\log\{1\} = 0$, because $\exp\{0\} = 1$ (Example 0.40).
- $\log\{e\} = 1$, because $\exp\{1\} = e$ (Definition 0.48).
- $\log\{0\}$ and $\log\{-2\}$ are not defined, because $\exp\{y\} > 0$ for every real y.

Definition 0.50 (Power with a real exponent)**.** For a real number $a > 0$ and a real number b,

$$a^b \stackrel{\text{def}}{=} \exp\{b \cdot \log\{a\}\}.$$

When b is an integer, this agrees with Definition 0.4 and Definition 0.5.

Example 0.43 (Real powers, and e^x)**.**

- $2^3 = \exp\{3 \cdot \log\{2\}\} \approx \exp\{3 \cdot 0.69315\} \approx \exp\{2.07944\} \approx 8$, which agrees with

$$\begin{aligned} 2^3 &= 2 \cdot 2 \cdot 2 \\ &= 8 \end{aligned}$$

  from Definition 0.4.

- $2^{1/2} = \exp\{\frac{1}{2} \cdot \log\{2\}\} \approx \exp\{0.34657\} \approx 1.41421$, which is $\sqrt{2}$ (Definition 0.6).
- For every real x, with base e (Definition 0.48):

$$\begin{aligned} e^x &= \exp\{x \cdot \log\{e\}\} && \text{(Definition 0.50, with } a = e \text{ and } b = x\text{)} \\ &= \exp\{x \cdot 1\} && (\log\{e\} = 1, \text{ by Example 0.42}) \\ &= \exp\{x\} && \text{(Theorem 0.13)} \end{aligned}$$

  So e^x and $\exp\{x\}$ are two names for the same number.

Definition 0.51 (Real powers of zero and of negative numbers)**.** Definition 0.4, Definition 0.5 and Definition 0.50 leave some powers b^c with $b \leq 0$ unassigned. We complete

them as follows.

- For a real number $c > 0$, $0^c \stackrel{\text{def}}{=} 0$. For a natural number c, this agrees with Definition 0.4.
- For $b < 0$, b^c is defined only when c is an integer, by Definition 0.4 and Definition 0.5; for $b < 0$ and a non-integer c, b^c is undefined.

Example 0.44 (Powers of zero and of negative numbers)**.**

- $0^{1/2} = 0$, which agrees with $\sqrt{0} = 0$ (Definition 0.6).
- $(-8)^2 = 64$ and $(-8)^{-1} = -\frac{1}{8}$ are defined, since the exponents are integers.
- $(-8)^{1/3}$ is undefined, even though $(-2)^3 = -8$. R follows the same convention: `(-8)^(1/3)` is `NaN`.

Theorem 0.22 (exp and log are mutual inverses)**.**

1. *For every* $a > 0$*:* $\exp\{\log\{a\}\} = a$.
2. *For every* $a \in \mathbb{R}$*:* $\log\{\exp\{a\}\} = a$.

Theorem 0.23 (Logarithm of a product)**.** *If* $a > 0$ *and* $b > 0$*, then*

$$\log\{a \cdot b\} = \log\{a\} + \log\{b\}$$

Corollary 0.1 (Logarithm of a quotient)**.** *If* $a > 0$ *and* $b > 0$*, then*

$$\log\left\{\frac{a}{b}\right\} = \log\{a\} - \log\{b\}$$

Proof

Proof. Since $a > 0$ and $b > 0$, the quotient $\frac{a}{b}$ is positive, so Theorem 0.23 applies to the product $\frac{a}{b} \cdot b$:

$$\begin{aligned} \log\{a\} &= \log\left\{\frac{a}{b} \cdot b\right\} && (a = \tfrac{a}{b} \cdot b) \\ &= \log\left\{\frac{a}{b}\right\} + \log\{b\} && \text{(logarithm of a product)} \end{aligned}$$

The second step applies Theorem 0.23. Subtracting $\log\{b\}$ from both sides gives $\log\{a\} - \log\{b\} = \log\{\frac{a}{b}\}$. □

Theorem 0.24 (Logarithm of a power)**.** *If $a > 0$ and $b \in \mathbb{R}$, then*

$$\log\{a^b\} = b \cdot \log\{a\}$$

Theorem 0.25 (Exponential of a sum)**.** *The exponential of a sum is equal to the product of the exponentials of its terms:*

$$\exp\{a + b\} = \exp\{a\} \cdot \exp\{b\}$$

Example 0.45 (Exponential of a sum)**.** With $a = 2$ and $b = 3$, $\exp\{2 + 3\} = \exp\{5\} \approx 148.41$, and $\exp\{2\} \cdot \exp\{3\} \approx 7.3891 \cdot 20.0855 \approx 148.41$.

Corollary 0.2 (Exponential of a difference)**.** *The exponential of a difference is the exponential of the first term divided by the exponential of the second term:*

$$\exp\{a - b\} = \frac{\exp\{a\}}{\exp\{b\}}$$

Example 0.46 (Exponential of a difference)**.** With $a = 5$ and $b = 2$, $\exp\{5 - 2\} = \exp\{3\} \approx 20.09$, and $\frac{\exp\{5\}}{\exp\{2\}} \approx \frac{148.4132}{7.3891} \approx 20.09$.

Theorem 0.26 (Powers of 1 and first powers)**.** *For every $b \in \mathbb{R}$,*

$$1^b = 1,$$

and for every $a \in \mathbb{R}$,

$$a^1 = a.$$

Theorem 0.27 (Power of a sum)**.** *If $a > 0$ and $b, c \in \mathbb{R}$, then*

$$a^{b+c} = a^b \cdot a^c$$

Example 0.47 (Power of a sum)**.** With $a = 2$, $b = 3$, and $c = 4$,

$$\begin{aligned} 2^{3+4} &= 2^7 \\ &= 128, \end{aligned}$$

and

$$\begin{aligned} 2^3 \cdot 2^4 &= 8 \cdot 16 \\ &= 128. \end{aligned}$$

Theorem 0.28 (Power of a product)**.** *If $a, b > 0$ and $c \in \mathbb{R}$, then*

$$(ab)^c = a^c \cdot b^c$$

When c is a positive integer, the same identity holds for all $a, b \in \mathbb{R}$, because both sides are products of c copies of a and c copies of b, which can be regrouped by Theorem 0.14 and Theorem 0.15.

Example 0.48 (Power of a product)**.** With $a = 2$, $b = 3$, and $c = 2$,

$$\begin{aligned} (2 \cdot 3)^2 &= 6^2 \\ &= 36, \end{aligned}$$

and

$$\begin{aligned} 2^2 \cdot 3^2 &= 4 \cdot 9 \\ &= 36. \end{aligned}$$

Theorem 0.29 (Power of a power)**.** *If $a > 0$ and $b, c \in \mathbb{R}$, then*

$$\begin{aligned} a^{bc} &= (a^b)^c \\ &= (a^c)^b \end{aligned}$$

Example 0.49 (A negative base)**.** With $a = -1$, $b = 2$, and $c = \frac{1}{2}$:

$$\begin{aligned} a^{bc} &= (-1)^{2 \cdot \frac{1}{2}} \\ &= (-1)^1 \\ &= -1 \end{aligned}$$

but

$$\begin{aligned} (a^b)^c &= ((-1)^2)^{\frac{1}{2}} \\ &= 1^{\frac{1}{2}} \\ &= 1 \end{aligned}$$

So $a^{bc} \neq (a^b)^c$ here, which is why Theorem 0.29 requires $a > 0$. The third expression, $(a^c)^b = \left((-1)^{\frac{1}{2}}\right)^2$, is not even a real number.

Corollary 0.3 (Natural exponential of a product)**.**

$$\begin{aligned}\exp\{ab\} &= (\exp\{a\})^b \\ &= (\exp\{b\})^a\end{aligned}$$

Remark (Tarski's high school identities)

Remark 0.18 (Tarski's high school identities). Restricted to positive integers, the following results are Tarski's eleven "high school" identities:

- sums are symmetric and associative (Theorem 0.11, Theorem 0.12);
- multiplying by 1 changes nothing, products are symmetric and associative, and multiplication is distributive (Theorem 0.13, Theorem 0.14, Theorem 0.15, Theorem 0.17);
- $1^b = 1$, $a^1 = a$, and the power of a sum, of a product, and of a power (Theorem 0.26, Theorem 0.27, Theorem 0.28, Theorem 0.29).

Remark (Tarski's identities are not complete)

Remark 0.19 (Tarski's identities are not complete). Tarski asked whether every identity in $+$, $\times$, exponentiation and 1 that is true for all positive integers can be derived from the eleven identities in Remark 0.18. It cannot: Wilkie found an identity that is true for all positive integers but does not follow from them. So this list is a useful core, not a complete rulebook.

Definition 0.52 (Contour line)**.** For a function g of two real variables b and c and a number k, the **contour line** of g at height k is the set of points where g equals k:

$$\{(b, c) : g(b, c) = k\}$$

Example 0.50 (Contour lines of a bowl)**.** For $g(b, c) = b^2 + c^2$, the contour line at height $k = 4$ is $\{(b, c) : b^2 + c^2 = 4\}$, the circle of radius 2 around $(0, 0)$: for example,

$$\begin{aligned}g(2, 0) &= 4 + 0 \\ &= 4\end{aligned}$$

and

$$\begin{aligned} g(0,-2) &= 0+4 \\ &= 4. \end{aligned}$$

The contour line at height $k=-1$ is empty, because $b^2+c^2 \geq 0$ for all b and c.

Remark (Contour lines on a plot of a surface)

Remark 0.20 (Contour lines on a plot of a surface). On a plot of the surface $z=g(b,c)$, the contour line is the curve along which the surface has height k.

Exercise 0.9 (Exponentiation versus multiplication)**.** For $b,c \in \mathbb{R}$, when does $b^c = bc$?

Solution

Solution 0.9. We only count a pair (b,c) when b^c is defined, so for $b<0$ we only consider integer c (Definition 0.51). With that convention, $bc=b^c$ in each of the following cases:

1. $c=1$, for every b.
2. $b=0$ and $c>0$, since then

$$\begin{aligned} b^c &= 0 \\ &= bc. \end{aligned}$$

   ($b=0$ and $c=0$ fails, since $0^0=1\neq 0$.)

3. $b>0$, $c>0$, $c\neq 1$, and $b=\exp\left\{\frac{\log\{c\}}{c-1}\right\}$. For $b>0$, dividing both sides of $b^c=bc$ by b gives $b^{c-1}=c$, which needs $c>0$ because $b^{c-1}>0$; taking logarithms then gives $(c-1)\log\{b\}=\log\{c\}$. For example, $c=2$ gives $b=2$, and indeed

$$\begin{aligned} 2^2 &= 4 \\ &= 2\cdot 2. \end{aligned}$$

4. $b<0$ and c is an odd integer with $c\geq 3$, with $b=-c^{1/(c-1)}$; for example, $b=-\sqrt{3}$ and $c=3$ give

$$\begin{aligned} b^c &= -3\sqrt{3} \\ &= bc. \end{aligned}$$

5. $b<0$ and c is an even integer with $c\leq -2$, with $b=-(-c)^{1/(c-1)}$; for example, $b=-2^{-1/3}$ and $c=-2$ give

$$b^c = 2^{2/3}$$
$$= bc.$$

For $b < 0$, cases 4 and 5 come from $b^{c-1} = c$ as well: when $c-1$ is even, $b^{c-1} > 0$, so c must be positive; when $c-1$ is odd, $b^{c-1} < 0$, so c must be negative.
See the red contours in Figure 2 for a visualization of the $b \geq 0$ cases.

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 1: Graph of $b * c$ and b^c

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 2: **Graph of** $b^c - b * c$. The red contour lines are at heights 0 and 1; the ones at height 0 show where $b^c = b * c$.

Theorem 0.30 (When exponentiation equals multiplication)**.** *Let $b, c \in \mathbb{R}$, where b^c is defined (for $b < 0$, only integer c is allowed; Definition 0.51). Then $b^c = bc$ if and only if one of the following holds:*

1. *$c = 1$;*
2. *$b = 0$ and $c > 0$;*
3. *$b > 0$, $c > 0$, $c \neq 1$, and $b = \exp\left\{\frac{\log\{c\}}{c-1}\right\}$;*
4. *$b < 0$, c is an odd integer with $c \geq 3$, and $b = -c^{1/(c-1)}$;*
5. *$b < 0$, c is an even integer with $c \leq -2$, and $b = -(-c)^{1/(c-1)}$.*

Proof

Proof. This is the solution to Exercise 0.9, worked in Solution 0.9. □

Exercise 0.10 (Repeated exponentiation)**.** For $a \geq 0$, $b, c \in \mathbb{R}$, when does $(a^b)^c = a^{(b^c)}$?

Solution

Solution 0.10. Short answer: rarely (that's all you need to know for this course).
Long answer:
Split on whether $a > 0$ or $a = 0$, because the logarithm we use for $a > 0$ is undefined at $a = 0$.
Case $a > 0$**.** By Theorem 0.29, $(a^b)^c = a^{bc}$, so the question becomes when $a^{bc} = a^{(b^c)}$ (for pairs (b, c) where b^c is defined). Because $a > 0$, both sides are positive, and we can

take logarithms (Theorem 0.24):

$$\begin{aligned} a^{bc} &= a^{(b^c)} \\ \log\{a^{bc}\} &= \log\{a^{(b^c)}\} \quad \text{(take logarithms of both sides)} \\ bc \cdot \log\{a\} &= b^c \cdot \log\{a\} \quad \text{(logarithm of a power)} \end{aligned} \tag{1}$$

The last line of Equation 1 holds exactly when

1. $a = 1$ (so that $\log\{a\} = 0$), or
2. $bc = b^c$ (see Theorem 0.30).

Case $a = 0$. Here we cannot take logarithms, so we work from the values of powers of 0: $0^s = 0$ for $s > 0$ (Definition 0.51), $0^0 = 1$, and 0^s is undefined for $s < 0$.

- If $b < 0$, then 0^b is undefined, so $(a^b)^c$ is undefined.
- If $b > 0$, then $(0^b)^c = 0^c$ and $b^c > 0$, so $0^{(b^c)} = 0$; the two sides agree exactly when $c > 0$.
- If $b = 0$, then

$$\begin{aligned} (0^0)^c &= 1^c \\ &= 1; \end{aligned}$$

  for $c > 0$,

$$\begin{aligned} 0^{(0^c)} &= 0^0 \\ &= 1, \end{aligned}$$

  so the two sides agree, and for $c \leq 0$ they do not.

So for $a = 0$, $(a^b)^c = a^{(b^c)}$ exactly when $b \geq 0$ and $c > 0$.
In particular, when $a = 0$, $b \geq 0$, and $c = 0$, the two sides differ:

$$\begin{aligned} (a^b)^c &= (0^b)^0 \\ &= 1 \end{aligned}$$

$$\begin{aligned} a^{(b^c)} &= 0^{(b^0)} \\ &= 0^1 \\ &= 0 \end{aligned}$$

Theorem 0.31 (When repeated exponentiation is not reordered)**.** *Let $a \geq 0$ and $b, c \in \mathbb{R}$, where b^c is defined. Then $(a^b)^c = a^{(b^c)}$, with both sides defined, if and only if*

- *$a > 0$ and either $a = 1$ or $bc = b^c$; or*
- *$a = 0$, $b \geq 0$, and $c > 0$.*

Proof

Proof. This is the solution to Exercise 0.10, worked in Solution 0.10. □

Cosine and sine

Definition 0.53 (Cosine and sine)**.** The *unit circle* is the set of points (x_1, x_2) in the plane with $x_1^2 + x_2^2 = 1$, and $\pi \approx 3.14159$ is half of its circumference. For a real number θ, start at the point $(1, 0)$ and move a distance $|\theta|$ along the unit circle, counterclockwise if $\theta \geq 0$ and clockwise if $\theta < 0$. The point reached is $(\cos\theta, \sin\theta)$: its first coordinate is the **cosine** of θ, and its second coordinate is the **sine** of θ.

Example 0.51 (Cosines and sines of some angles)**.**

- $\theta = 0$ stays at $(1, 0)$, so $\cos 0 = 1$ and $\sin 0 = 0$.
- $\theta = \pi$ goes half way around the circle, to $(-1, 0)$, so $\cos\pi = -1$ and $\sin\pi = 0$.
- $\theta = \pi/2$ goes a quarter of the way round, to $(0, 1)$, so $\cos(\pi/2) = 0$ and $\sin(\pi/2) = 1$.
- $\theta = \pi/4$ goes an eighth of the way round, to the point of the circle with $x_1 = x_2 > 0$. There $x_1^2 + x_1^2 = 1$, so $x_1^2 = \frac{1}{2}$ and $x_1 = \frac{1}{\sqrt{2}} \approx 0.7071$:

$$\begin{aligned} \cos(\pi/4) &= \sin(\pi/4) \\ &= \tfrac{1}{\sqrt{2}}. \end{aligned}$$

Angles are also measured in *degrees*, with a full turn of 2π equal to 360 degrees, so $\pi/2$ is 90 degrees and $\pi/4$ is 45 degrees.

Theorem 0.32 (The Pythagorean identity)**.** *For every real number θ,*

$$\cos^2\theta + \sin^2\theta = 1.$$

Proof

Proof. The point $(\cos\theta, \sin\theta)$ lies on the unit circle (Definition 0.53), so its coordinates satisfy $x_1^2 + x_2^2 = 1$. □

Example 0.52 (The unit circle and the angle $\pi/3$)**.** Take $\theta = \pi/3$, one sixth of the way round the circle. Figure 3 shows the point reached, $(\cos\theta, \sin\theta)$, and its two coordinates. The point is a vertex of a regular hexagon inscribed in the unit circle, whose sides have the same length as its radius, so the point is at distance 1 from $(1, 0)$. Writing $(x_1, x_2) = (\cos\theta, \sin\theta)$,

$$\begin{aligned} 1 &= (x_1 - 1)^2 + x_2^2 \\ &= x_1^2 + x_2^2 - 2x_1 + 1 \\ &= 2 - 2x_1, \end{aligned}$$

so $\cos(\pi/3) = x_1 = \frac{1}{2}$, and then $\sin(\pi/3) = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$. To three decimal places these are 0.5 and 0.866. As Theorem 0.32 says, the squared coordinates add to 1.

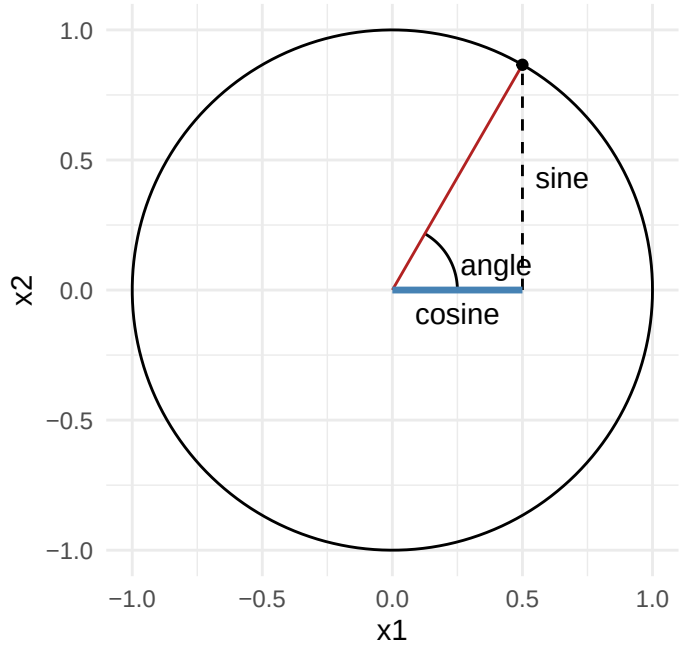


Figure 3: The unit circle with the point at angle $\pi/3$. Its horizontal coordinate is the cosine and its vertical coordinate is the sine.

Theorem 0.33 (Cosine takes each value from -1 to 1 exactly once between 0 and π)**.** *The function* $\cos : [0, \pi] \to [-1, 1]$ *is decreasing, with* $\cos 0 = 1$ *and* $\cos \pi = -1$*, and takes each value in* $[-1, 1]$ *exactly once. On* $[0, \pi]$*,* $\sin \theta \geq 0$*.*

Proof

Proof. For $\theta \in [0, \pi]$, the point $(\cos\theta, \sin\theta)$ is at distance θ along the circle from $(1, 0)$, counterclockwise, so it lies on the upper half of the unit circle, $\{(x_1, x_2) : x_1^2 + x_2^2 = 1,\ x_2 \geq 0\}$, which gives $\sin\theta \geq 0$. Different values of θ in $[0, \pi]$ are different distances along that half circle, so they give different points. Two different points of the upper half have different first coordinates, because $x_2 = \sqrt{1 - x_1^2}$ is determined by x_1. So cos is one-to-one on $[0, \pi]$. Every $x_1 \in [-1, 1]$ is the first coordinate of the point $(x_1, \sqrt{1 - x_1^2})$ of the upper half, and the distance along the half circle from $(1, 0)$ to that point is some $\theta \in [0, \pi]$, so cos takes every value in $[-1, 1]$. Finally, move along the upper half circle from $(1, 0)$ to $(-1, 0)$ as θ increases from 0 to π. The first coordinate never takes the same value twice, so it never turns back. It starts at $\cos 0 = 1$ and ends at $\cos \pi = -1$, so it only decreases. □

Complex numbers

Definition 0.54 (Imaginary unit)**.** The **imaginary unit** i is a number whose square is -1:

$$i^2 \stackrel{\text{def}}{=} -1.$$

Apart from that one rule, i obeys the usual rules of arithmetic:

- sums and products with i are commutative;
- sums and products with i are associative;
- multiplication distributes over addition.

Axler (2024, Definition 1.1, p. 2) makes this idea rigorous: it defines a complex number as an ordered pair (a, b) of real numbers, written $a + bi$, defines addition and multiplication of such pairs, writes $0 + 1i$ as i, and leaves it to the reader to verify that $i^2 = -1$.

Example 0.53 (Powers of the imaginary unit)**.** The powers of i repeat in a cycle of four:

$$\begin{aligned} i^3 &= i^2 \cdot i && \text{(split off one factor of } i\text{)} \\ &= (-1) \cdot i && \text{(Definition 0.54)} \\ &= -i && \text{(multiply)} \end{aligned}$$

and

$$\begin{aligned} i^4 &= i^2 \cdot i^2 && \text{(split the power into two squares)} \\ &= (-1) \cdot (-1) && \text{(Definition 0.54, twice)} \\ &= 1 && \text{(multiply)} \end{aligned}$$

so the cycle starts again:

$$\begin{aligned} i^5 &= i^4 \cdot i && \text{(split off one factor of } i\text{)} \\ &= 1 \cdot i && (i^4 = 1) \\ &= i && \text{(multiply)} \end{aligned}$$

and the powers run $i, -1, -i, 1, i, \ldots$.

Example 0.54 (No real number squares to -1)**.** The imaginary unit is not a real number, because the square of every real number x is at least 0:

- if $x \geq 0$, then $x^2 = x \cdot x$ is a product of two nonnegative numbers, so $x^2 \geq 0$;
- if $x < 0$, then $x^2 = x \cdot x$ is a product of two negative numbers, so $x^2 > 0$.

For instance, $3^2 = 9$ and $(-3)^2 = 9$; neither is -1. So the equation $x^2 = -1$ has no real solution, and Definition 0.54 introduces a new number to solve it.

Theorem 0.34 (The number $-i$ also squares to -1)**.**

$$(-i)^2 = -1.$$

Proof

Proof.

$$\begin{aligned} (-i)^2 &= (-1)^2\, i^2 && (-i = (-1) \cdot i, \text{ and } (ab)^2 = a^2 b^2 \text{ because products commute and associate}) \\ &= 1 \cdot (-1) && ((-1)^2 = 1 \text{ and Definition 0.54}) \\ &= -1 && \text{(multiply)} \end{aligned}$$

□

Example 0.55 (Squaring i and $-i$ in R)**.** R has the imaginary unit as `1i`. The code below squares it and its negative.

```
squares <- c(i = (1i)^2, minus_i = (-1i)^2)
squares
#>       i minus_i
#> -1+0i -1+0i
```

Both squares have real part -1 and imaginary part 0, as Definition 0.54 and Theorem 0.34 say.

Remark (The rule does not tell i from $-i$)

Remark 0.21 (The rule does not tell i from $-i$). Because $-i$ obeys the same rule as i (Theorem 0.34), the rule $i^2 = -1$ does not single one of them out. The symbol i names one of these two square roots of -1, chosen once and for all.

Definition 0.55 (Complex number)**.** A **complex number** is a number of the form

$$z = a + b\,i,$$

where a and b are real numbers and i is the imaginary unit (Definition 0.54).

- The **real part** of z is $\operatorname{Re} z \stackrel{\text{def}}{=} a$.
- The **imaginary part** of z is $\operatorname{Im} z \stackrel{\text{def}}{=} b$.
- The set of all complex numbers is $\mathbb{C} \stackrel{\text{def}}{=} \{a + b\,i : a, b \in \mathbb{R}\}$.

Two complex numbers are equal exactly when their real parts are equal and their imaginary parts are equal.

Axler (2024, Definition 1.1, p. 2) defines $\mathbb{C}$ as the set of ordered pairs (a, b) of real numbers, written $a + bi$, and Axler (2024, Definition 4.1, p. 120) defines the real and imaginary parts.

Example 0.56 (Real and imaginary parts)**.** For $z = 2 - 5\,i$, the real part is $\operatorname{Re} z = 2$ and the imaginary part is $\operatorname{Im} z = -5$: the imaginary part is the real number -5, not $-5\,i$.
The numbers $1 + i$ and $1 - i$ have the same real part, 1, but different imaginary parts, 1 and -1, so they are different complex numbers.
The real number 7 is the complex number $7 + 0\,i$, with real part 7 and imaginary part 0.

Theorem 0.35 (Every real number is a complex number)**.** *A complex number z is a real number exactly when* $\operatorname{Im} z = 0$*. In particular, every real number a is the complex number $a + 0\,i$, with* $\operatorname{Re} a = a$ *and* $\operatorname{Im} a = 0$*, so $\mathbb{R}$ is a subset of $\mathbb{C}$.*

Proof

Proof. Let a be a real number. Since $0\,i = 0$,

$$\begin{aligned} a + 0\,i &= a + 0 && (0\,i = 0) \\ &= a && \text{(Theorem 0.10)} \end{aligned}$$

so a is the complex number $a + 0\,i$, whose imaginary part is 0. The real part of $a + 0\,i$ is a, and by the equality rule in Definition 0.55 no other pair of real and imaginary parts gives the same number.
Conversely, if $z = a + b\,i$ has $b = \operatorname{Im} z = 0$, then $z = a + 0\,i = a$ by the same calculation, so z is real. □

Example 0.57 (Which complex numbers are real)**.** R stores complex numbers as a real part and an imaginary part. The code below converts the real number 7 to a complex number and reads the parts of three complex numbers.

```
z <- c(as.complex(7), 2 - 5i, 3 + 0i)
data.frame(z = z, re = Re(z), im = Im(z), is_real = Im(z) == 0)
#>      z re im is_real
#> 1 7+0i  7  0    TRUE
#> 2 2-5i  2 -5   FALSE
#> 3 3+0i  3  0    TRUE
```

The real number 7 becomes the complex number 7+0i, with imaginary part 0. Of the 3 numbers, 2 have imaginary part 0, and by Theorem 0.35 these are the real ones. The number $2 - 5\,i$ is not real, because its imaginary part is -5.

Theorem 0.36 (Adding and multiplying complex numbers)**.** *For real numbers a, b, c, d:*

$$(a + b\,i) + (c + d\,i) = (a + c) + (b + d)\,i$$

and

$$(a + b\,i)(c + d\,i) = (ac - bd) + (ad + bc)\,i.$$

Proof

Proof. **Sum.**

$$\begin{aligned}(a+b\,i)+(c+d\,i) &= a+c+b\,i+d\,i && \text{(rearrange the terms)}\\ &= (a+c)+(b+d)\,i && \text{(factor out } i\text{)}\end{aligned}$$

Product.

$$\begin{aligned}(a+b\,i)(c+d\,i) &= ac+ad\,i+bc\,i+bd\,i^2 && \text{(distribute)}\\ &= ac+ad\,i+bc\,i-bd && \text{(Definition 0.54)}\\ &= (ac-bd)+(ad+bc)\,i && \text{(group the real terms and the terms with } i\text{)}\end{aligned}$$

□

Axler (2024, Definition 1.1, p. 2) takes these two formulas as the definitions of addition and multiplication.

Example 0.58 (Adding and multiplying two complex numbers)**.** Let $w = 1+2\,i$ and $z = 3-i$. Their sum is

$$\begin{aligned}w+z &= (1+3)+(2+(-1))\,i && \text{(Theorem 0.36, sum)}\\ &= 4+i && \text{(add)}\end{aligned}$$

and their product, multiplying out directly, is

$$\begin{aligned}wz &= 1\cdot 3+1\cdot(-i)+2\,i\cdot 3+2\,i\cdot(-i) && \text{(distribute)}\\ &= 3-i+6\,i-2\,i^2 && \text{(multiply)}\\ &= 3-i+6\,i+2 && \text{(Definition 0.54)}\\ &= (3+2)+(-1+6)\,i && \text{(group the real terms and the terms with } i\text{)}\\ &= 5+5\,i && \text{(add)}\end{aligned}$$

The product formula in Theorem 0.36 gives the same answer: with $a=1$, $b=2$, $c=3$ and $d=-1$,

$$\begin{aligned}ac-bd &= 3-(-2)\\ &= 5\end{aligned}$$

and

$$\begin{aligned}ad+bc &= -1+6\\ &= 5.\end{aligned}$$

Definition 0.56 (Complex conjugate)**.** The **complex conjugate** of a complex number $z = a + b\,i$ (Definition 0.55) is

$$\overline{z} \stackrel{\text{def}}{=} a - b\,i.$$

Axler (2024, Definition 4.2, p. 120) gives the same definition, as $\overline{z} = \operatorname{Re} z - (\operatorname{Im} z)\,i$.

Example 0.59 (Complex conjugates)**.**

- $\overline{3 + 4\,i} = 3 - 4\,i$.
- $$\begin{aligned}\overline{-2\,i} &= \overline{0 + (-2)\,i} \\ &= 0 - (-2)\,i \\ &= 2\,i.\end{aligned}$$
- $$\begin{aligned}\overline{5} &= \overline{5 + 0\,i} \\ &= 5 - 0\,i \\ &= 5\,:\end{aligned}$$

  a real number is its own complex conjugate. A complex number with a nonzero imaginary part, such as $3 + 4\,i$, is not.

Definition 0.57 (Absolute value (modulus) of a complex number)**.** The **absolute value**, or **modulus**, of a complex number $z = a + b\,i$ (Definition 0.55) is

$$|z| \stackrel{\text{def}}{=} \sqrt{a^2 + b^2}.$$

Axler (2024, Definition 4.2, p. 120) gives the same definition.

Example 0.60 (Absolute values of complex numbers)**.**

- $$\begin{aligned}|3 + 4\,i| &= \sqrt{3^2 + 4^2} \\ &= \sqrt{25} \\ &= 5.\end{aligned}$$
- $$\begin{aligned}|-2\,i| &= \sqrt{0^2 + (-2)^2} \\ &= \sqrt{4} \\ &= 2.\end{aligned}$$

- For a real number $a = a + 0\,i$,

$$\begin{aligned} |a| &= \sqrt{a^2 + 0^2} \\ &= \sqrt{a^2}, \end{aligned}$$

which is the usual absolute value of a; for instance,

$$\begin{aligned} |-3| &= \sqrt{9} \\ &= 3. \end{aligned}$$

Theorem 0.37 (A complex number times its conjugate)**.** *For every complex number z,*

$$z\,\overline{z} = |z|^2.$$

In particular, $z\,\overline{z}$ is a nonnegative real number.

Proof

Proof. Write $z = a + b\,i$ with a, b real. Then

$$\begin{aligned} z\,\overline{z} &= (a + b\,i)(a - b\,i) && \text{(Definition 0.56)} \\ &= a^2 - ab\,i + ab\,i - b^2\,i^2 && \text{(distribute)} \\ &= a^2 - b^2\,i^2 && \text{(cancel } ab\,i\text{)} \\ &= a^2 + b^2 && \text{(Definition 0.54)} \\ &= |z|^2 && \text{(Definition 0.57)} \end{aligned}$$

□

Axler (2024, result 4.4, p. 121) lists this identity among the properties of complex numbers.

Example 0.61 (Multiplying $3 + 4\,i$ by its conjugate)**.**

$$\begin{aligned} &(3 + 4\,i)(3 - 4\,i) \\ &= 9 - 12\,i + 12\,i - 16\,i^2 && \text{(distribute)} \\ &= 9 - 16\,i^2 && \text{(cancel } 12\,i\text{)} \\ &= 9 + 16 && \text{(Definition 0.54)} \\ &= 25 && \text{(add)} \end{aligned}$$

which is $|3 + 4\,i|^2 = 5^2$ from Example 0.60, as Theorem 0.37 says.

Further reading

- Abramson et al. (2021) is a free online textbook on algebra and trigonometry. It covers equations, inequalities, polynomials, exponentials, logarithms, and trigonometry, which overlap with the algebra in these notes.
- Rudin (1976) develops the real numbers, including the least upper bound property behind the infimum and supremum.

References

Abramson, Jay et al. 2021. *Algebra and Trigonometry*. 2nd ed. OpenStax. https://openstax.org/books/algebra-and-trigonometry-2e/pages/preface.

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